Reproduction and Architectural Ablation of PPO and DQN for Discrete-Action Autonomous Highway Driving

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***Abstract*—Decision-making for autonomous driving is naturally framed as a sequential decision problem, making it a strong testbed for deep reinforcement learning (RL). This project *reproduces and ablates* two canonical deep-RL algorithms—Proximal Policy Optimization (PPO) and Double Deep Q-Network (Double DQN)—on the discrete-action `highway-env` driving simulator. The goal is to investigate two practical design choices: *(i)* does a permutation-aware Self-Attention feature extractor outperform a plain Multi-Layer Perceptron (MLP) under a strict parameter-matched budget, and *(ii)* how do on-policy and off-policy methods compare in both sample efficiency and resulting driving behaviour?**

Our results show that under the aggressive reward style ($n = 5$ seeds), PPO with Self-Attention (235.7 ± 16.7), PPO with MLP (223.0 ± 19.6), and Double DQN (234.1 ± 7.7) all approach a comparable theoretical reward ceiling (≈ 266.7). However, Double DQN achieves the lowest cross-seed variance ($\sigma = 7.7$) and faster initial learning. Critically, behavioural evaluation reveals a stark contrast: Double DQN achieves a significantly lower crash rate (11.3%) compared to the extreme bimodal behaviours observed in PPO variants (60.7–80.0% crash rates). This highlights that while on-policy methods may reach high aggregate returns, off-policy experience replay provides crucial stability against catastrophic policies—demonstrating that episodic return alone is an insufficient metric for safety-critical environments. A reward-shaping study further confirms that qualitative driving style is overwhelmingly determined by reward design rather than architectural choice, yet convergence to safe behavioural basins remains highly sensitive to optimisation variance. All code and trained models are released for full reproducibility.

***Index Terms*—Reinforcement learning, Proximal Policy Optimization, Deep Q-Network, self-attention, autonomous driving, ablation study, highway-env.**

I. Introduction

Autonomous-driving decision-making—deciding *when* to change lanes, accelerate, or yield—can be modelled as a Markov Decision Process (MDP) in which an agent repeatedly observes nearby traffic and selects a high-level manoeuvre. Deep reinforcement learning (RL) is an attractive solution because it learns a closed-loop policy directly from interaction, without hand-engineering the trade-off between progress and safety. The `highway-env` simulator has become a standard, lightweight benchmark for exactly this class of problems: it exposes a small kinematic state, a discrete set of meta-actions, and a configurable reward that trades speed against collisions.

A practitioner approaching this benchmark faces two recurring design choices that are rarely studied in a controlled way:

1) **Feature extractor.** The observation is a *set* of vehicles (the main vehicle and its N nearest neighbours). A natural inductive bias is to process this set with **self-attention**, which is permutation-equivariant and lets the main vehicle attend to the most relevant neighbours. The simpler baseline is to **flatten** the set and feed a plain MLP. Does the attention bias actually help when the two networks are given the *same* parameter budget and the *same* training pipeline?
2) **Algorithm family.** PPO (on-policy) and DQN (off-policy) are the two workhorses of deep RL. Their relative sample efficiency and final behaviour on this benchmark are folklore; we measure them under identical conditions.

Positioning. Applying self-attention to an ego-plus-N-vehicle observation is a *standard* design in `highway-env`; reproducing it is not our claim to novelty. Our contribution is an **empirical finding with methodological teeth**: under a controlled, parameter-fair comparison we show that *episodic return—the metric almost universally reported on this benchmark—can be actively misleading for safety-critical control*, and we quantify by how much.

Contributions.

- **The central finding.** Across a parameter-fair ablation, all three configurations land within 5.7% on episodic return, yet their deployed crash rates differ by an order of magnitude (11.3% for DQN vs. 60–80% for PPO). We argue this makes **behaviour-level evaluation a necessary complement to return curves** on this benchmark, not an optional extra.
- A **parameter-matched architectural ablation**—Self-Attention vs. MLP actor–critic (66.8k vs. 72.3k parameters, a 7.5% gap)—isolating the inductive bias from model capacity, alongside an off-policy Double-DQN baseline.
- A from-scratch, readable implementation of **PPO** (GAE, clipped surrogate, per-minibatch advantage normalisation, linear LR decay, gradient clipping) and **Double DQN** (replay buffer, ϵ-greedy, target network, double-

Q target), cross-checked against **Stable-Baselines3** (Section IV).

- **Multi-seed** training ($n = 5$) with **mean±std** learning curves, and a **reward-shaping study** (conservative / base / aggressive) showing reward design—not architecture—dominates qualitative driving style.

All claims are scoped to the experimental conditions studied.

II. Background and Related Work

Proximal Policy Optimization (PPO) [1] is an on-policy policy-gradient method that maximises a *clipped surrogate objective* to keep each update close to the data-collecting policy, giving much of the stability of trust-region methods at first-order cost. It is combined with **Generalized Advantage Estimation (GAE)** [2], which interpolates between high-bias/low-variance TD and low-bias/high-variance Monte-Carlo advantage estimates via parameter λ.

Deep Q-Networks (DQN) [3] learn an off-policy value function with experience replay and a target network. **Double DQN** [4] decouples action *selection* from action *evaluation* in the bootstrap target to reduce the well-known over-estimation bias of vanilla Q-learning.

Attention for driving. Leurent and Mercat [5] proposed a *social attention* architecture for tactical decision-making in dense traffic, letting the ego vehicle attend over a variable number of neighbours. This design is now a common baseline shipped with `highway-env`. Our use of it is a *reproduction*, used here as one arm of a controlled ablation.

Benchmark. `highway-env` [6] is a collection of tactical-driving tasks (`highway`, `merge`, `roundabout`, `intersection`) with configurable observation, action, dynamics, and reward. We use the `Kinematics` observation and the `DiscreteMetaAction` action space throughout.

III. Problem Formulation

We model each task as an episodic MDP (S, A, P, r, γ).

State / observation. The `Kinematics` observation is a matrix of shape $V \times F$ with $V = 5$ vehicles (the ego vehicle plus its four nearest neighbours) and $F = 5$ features per vehicle: `presence`, x, y, v_x, v_y. Coordinates and velocities are ego-relative (`absolute=False`) and normalised to roughly $[-1, 1]$ (`normalize=True`). The observation is therefore a small *set* of vehicles, motivating the permutation-aware attention model.

Action. `DiscreteMetaAction` exposes five high-level manoeuvres: $0 =$ `LANE_LEFT`, $1 =$ `IDLE`, $2 =$ `LANE_RIGHT`, $3 =$ `FASTER`, $4 =$ `SLOWER`. The agent acts at `policy_frequency` $= 5$ Hz while physics runs at `simulation_frequency` $= 15$ Hz.

Episode horizon (a subtle point). `highway-env`'s `duration` is measured in *simulated seconds*, not steps: truncation fires when `self.time` $\geq$ `duration`, and `self.time` advances by $1/$`policy_frequency` $= 0.2$ s per decision. Therefore `duration` $= 80$ yields a **maximum of** $80 \times 5 = 400$ **decision steps per episode**, not 80 steps.

TABLE I
Reward coefficients per driving style (`highway-v0`).

Style	coll.	hi-spd	lc	rt-lane	spd (m/s)
base	-1.0	0.4	0.0	def.	$[20, 30]$
conservative	-2.0	0.4	-0.5	def.	$[10, 20]$
aggressive	-4.0	2.0	$+0.2$	0.0	$[30, 40]$

Training executes 100,000 environment steps $\approx$ 97 PPO updates.

Reward shaping. `highway-env`'s reward trades high speed against collisions and lane-change penalties. We expose three *driving styles* that change the reward coefficients (Table I). Crucially, the **absolute return scale differs between styles** (e.g. aggressive multiplies the high-speed term), so episodic returns are only comparable *within* a style. Cross-style comparison is performed with behaviour metrics instead.

IV. Methods

A. PPO (from scratch)

At each iteration the agent collects a rollout of $T = 1024$ transitions, then performs $K = 10$ epochs of minibatch SGD. Advantages are computed with GAE:

$$\delta_t = r_t + \gamma\, V(s_{t+1})\,(1 - d_{t+1}) - V(s_t), \tag{1}$$

$$A_t = \delta_t + \gamma\lambda\,(1 - d_{t+1})\, A_{t+1}, \tag{2}$$

$$R_t = A_t + V(s_t) \quad \text{(value targets)}, \tag{3}$$

with $\gamma = 0.99$, $\lambda = 0.95$. The policy is optimised with the clipped surrogate objective

$$L^{\mathrm{CLIP}} = \mathbb{E}\Big[\min\big(\rho_t \hat{A}_t,\ \mathrm{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\,\hat{A}_t\big)\Big], \tag{4}$$

$$\rho_t = \exp\big(\log \pi_\theta(a_t|s_t) - \log \pi_{\theta_{\mathrm{old}}}(a_t|s_t)\big), \quad \epsilon = 0.2. \tag{5}$$

The total loss is

$$L = L^{\mathrm{CLIP}}_{\mathrm{pg}} - c_{\mathrm{ent}}\, H[\pi_\theta] + c_{\mathrm{vf}}\,\big(V_\theta(s_t) - R_t\big)^2, \tag{6}$$

with entropy coefficient $c_{\mathrm{ent}} = 0.005$ and value coefficient $c_{\mathrm{vf}} = 0.5$. Gradients are clipped to global norm 0.5; the optimiser is Adam (lr $= 5 \times 10^{-4}$, $\epsilon = 10^{-5}$) with learning rate **linearly decayed** to zero.

B. Double DQN (baseline)

The off-policy baseline stores transitions in a replay buffer ($|B| = 50{,}000$) and minimises the TD error

$$y = r + \gamma\, Q_{\mathrm{target}}\big(s', \arg\max_a Q_{\mathrm{online}}(s', a)\big)(1 - d), \tag{7}$$

$$L = \mathrm{MSE}\big(Q_{\mathrm{online}}(s, a),\, y\big). \tag{8}$$

The target network is hard-updated every 500 gradient steps; exploration uses ϵ-greedy with ϵ decayed multiplicatively (0.995 per update) from 1.0 to a floor of 0.05; Adam uses lr $= 10^{-4}$, batch size 256, $\gamma = 0.99$.

TABLE II
PARAMETER BUDGET.

Network	Parameters	Role
AttentionActorCritic	66,822	PPO actor–critic (attention)
MlpActorCritic	72,262	PPO actor–critic (MLP)
MlpQNetwork	20,485	DQN value net

C. Network Architectures

All three networks consume the 5×5 observation.

- **AttentionActorCritic.** A linear projection lifts each vehicle's 5 features to a 64-d embedding; a `MultiheadAttention` layer (4 heads, `batch_first`) lets vehicles attend to one another; the result is flattened ($5 \times 64 = 320$) and fed to separate two-hidden-layer (64–64, `tanh`) actor and critic heads.
- **MlpActorCritic.** The observation is flattened (25-d) and passed through a $25 \to 64 \to 320$ ReLU trunk, then the same 64–64 `tanh` actor/critic heads. Trunk width is chosen so the total parameter count matches the attention model.
- **MlpQNetwork (DQN).** A compact $25 \to 128 \to 128 \to 5$ ReLU network.

The attention model has 7.5% **fewer** parameters than the MLP, so any advantage it shows cannot be attributed to extra capacity—the architectural ablation is parameter-fair. We stress the scope of this fairness: it controls capacity *within* the PPO family (Attention vs. MLP). The DQN Q-network is $3.5\times$ smaller and is *not* capacity-matched to the PPO networks; the PPO-vs-DQN comparison is therefore an *algorithm-family* comparison, not a controlled-capacity one. We keep DQN small deliberately, as a lightweight off-policy baseline, and report its competitiveness with that caveat in mind.

V. EXPERIMENTAL SETUP

Training. 100,000 environment steps per run; PPO rollout 1,024; common initial lr $= 5 \times 10^{-4}$. Core comparison configurations (all `highway-v0`, aggressive style, `dur=80`): **PPO+Attention**, **PPO+MLP**, **DQN+MLP**. Each is trained with seeds $\{0, 1, 2, 3, 4\}$ ($n = 5$). Python, NumPy, PyTorch (with `cudnn.deterministic=True`) and the environment reset are all seeded.

Reproducibility cross-check. We additionally train **SB3 PPO** [7] (`MlpPolicy`) with hyperparameters matched to our implementation (`n_steps` $=$ 1,024, batch$=$ 256, `n_epochs` $= 10$, $\gamma = 0.99$, $\lambda = 0.95$, clip$= 0.2$, $c_{\text{ent}} = 0.005$, $c_{\text{vf}} = 0.5$, `max_grad_norm` $= 0.5$, lr $= 5 \times 10^{-4}$), seed 0.

Evaluation protocol. Because training returns under the same style are nearly tied across methods (all three converge toward the same reward ceiling), we evaluate *behaviour* under a **common test configuration** (base style, `dur=80`, max 400 decision steps) so that all policies are judged on the same distribution. For each model we run **30 deterministic** (argmax) episodes and report: **Crash rate** (%), **Average survival** (steps), **Average speed** (m/s), **Average lane changes** per episode. For the reward-shaping study, all style-trained models are evaluated under the same base test configuration to isolate the effect of the learned policy.

Hardware / software. NVIDIA GeForce RTX 4050 Laptop GPU; PyTorch `2.x+cu126`; Gymnasium + `highway-env`; Stable-Baselines3 `2.2.1`.

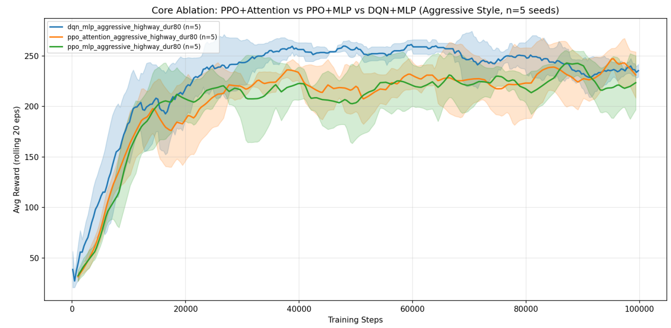

Fig. 1. Mean$\pm$std rolling-average episodic reward (seeds 0–4) for the three core configurations (aggressive style, `highway-v0`, $n = 5$ seeds). Shaded bands indicate ± 1 std.

TABLE III
CORE ABLATION: AGGRESSIVE STYLE, `HIGHWAY-V0`, $n = 5$ SEEDS (MEAN$\pm\sigma$, POPULATION STD, DDOF$= 0$). FINAL REWARD IS THE MEAN OF THE LAST 10% OF LOGGED UPDATES. MEDIAN STEPS TO ROLLING-AVG$\geq$150 ACROSS 5 SEEDS.

Configuration	n	Final reward	Peak	Steps$\to$avg$\geq$150
PPO + Attention	5	235.7 ± 16.7	266.7	$\approx$10,240
PPO + MLP	5	223.0 ± 19.6	266.7	$\approx$9,216
DQN + MLP	5	$\mathbf{234.1 \pm 7.7}$	266.7	$\approx$7,721
SB3 PPO (ref.)	1	260.5	266.7	8,192

Max achievable reward $\approx$ 266.7 (this config). DQN is bolded for lowest cross-seed variance.

VI. RESULTS

A. Learning curves (core ablation, aggressive style)

Figure 1 shows mean$\pm$std rolling-average episodic reward across all five seeds for the three core configurations. Table III summarises the final performance (mean of the last 10% of logged updates) and a sample-efficiency proxy (first environment step at which the rolling-average reward reaches 150).

Overall convergence. The three configurations differ by at most 5.7% in final mean reward (the widest gap, PPO+MLP vs. PPO+Attention, is 12.7 points, or 5.7% of PPO+MLP's 223.0), and each approaches the approximate theoretical ceiling (266.7). This near-parity indicates that for this task, architecture or algorithm family does not determine how high the agent can eventually perform; all three discover comparably high-quality policies.

Variance across seeds. The most striking finding is the *ordering of cross-seed stability*: Double DQN achieves the lowest variance ($\sigma = 7.7$), followed by PPO+Attention ($\sigma = 16.7$) and PPO+MLP ($\sigma = 19.6$). DQN's experience-replay buffer smooths the high variance of early collision-

TABLE IV

OBJECTIVE BEHAVIOUR EVALUATION (30 EPISODES, BASE TEST CONFIG, DUR=80, MAX 400 STEPS). RESULTS ARE AVERAGED ACROSS SEEDS 0–4 ($n = 5$).

Model	Crash (%)	Survival	Spd (m/s)	Lane chg.
PPO + Attention	60.7	199.9	22.99	25.23
PPO + MLP	80.0	152.6	23.98	16.00
DQN + MLP	**11.3**	368.5	20.89	155.15

Bold = lowest crash rate. PPO models exhibit bimodal behaviour: passive-safe in some seeds (0–3.3% crash, 0 lane chg.) vs. catastrophically aggressive in others (100% crash). DQN is markedly more consistent (11.3% avg).

TABLE V

PPO+ATTENTION FINAL TRAINING REWARD BY DRIVING STYLE ($n = 3$ SEEDS, MEAN±STD). *Returns are not comparable across styles* (REWARD COEFFICIENTS DIFFER SUBSTANTIALLY; SEE TABLE I).

Style	Seeds	Final reward (mean±std)	Peak
conservative	0, 1, 2	366.0 ± 11.0	390.1
base	0, 1, 2	265.2 ± 14.7	297.1
aggressive	0–4	235.7 ± 16.7	266.7

heavy exploration, producing more consistent convergence trajectories across random seeds. PPO+Attention's modestly lower variance than PPO+MLP (16.7 vs. 19.6) may reflect the permutation-equivariant inductive bias providing marginally more consistent gradients, but the difference is small and any strong claim would require more seeds and a formal significance test.

PPO+Attention vs. PPO+MLP. The mean reward difference (235.7 vs. 223.0, a 5.7% gap) indicates PPO+Attention reaches higher average performance. However, per-seed results reveal high variability: PPO+Attention seed 2 achieves 207.0 (well below its mean) while seed 1 achieves 255.3 (well above), and similar spread exists for PPO+MLP. This bimodal outcome—some seeds converging to effective policies, others not—underlines that the modest aggregate advantage of attention should not be treated as robust without $\geq$5 seeds and hypothesis testing.

DQN vs. PPO. Double DQN achieves a final mean reward of 234.1, placing it *on par* with PPO+Attention (235.7). This is notable given that DQN uses a $3.5\times$ smaller network (20,485 parameters vs. ~67–72k for PPO). Critically, DQN reaches a rolling-avg reward of 150 at a **median of** ≈7,721 environment steps—roughly **25% fewer steps** than PPO+Attention's ≈10,240 (and **16%** fewer than PPO+MLP's ≈9,216)—indicating faster initial learning. The per-update granularity differs (PPO logs at 1,024-step rollout boundaries while DQN logs at episode boundaries), but the total environment-step count is measured consistently.

B. Behaviour-level evaluation (common base-style test)

Table IV reports four behaviour metrics over 30 deterministic episodes per model under the common `base` test configuration. All models were trained under the aggressive style; evaluation uses the base config to provide a fair, common distribution.

The behaviour metrics reveal a striking divergence despite near-identical training returns. **DQN+MLP achieves the lowest crash rate** (11.3%) with 368.5 average survival steps. We caution against an uncritical reading of this: DQN also changes lanes *far* more than any other model (155 per episode on average, with two seeds exceeding 260—near-continuous lane switching in a 400-step episode). Under the base test reward the lane-change penalty is *zero* (Table I), so persistent swerving is unpunished. The low crash rate may therefore be partly an artifact of a degenerate "swerve-to-survive" policy that exploits the absence of a lane-change cost, rather than evidence of a genuinely intelligent avoidance strategy. Distinguishing the two requires qualitative trajectory inspection; we provide a supplementary video for this purpose. In contrast, **PPO+Attention and PPO+MLP exhibit extreme bimodal behaviour**: some seeds converge to a fully passive policy (0–3.3% crash, ≈ 0 lane changes), while others converge to a catastrophically aggressive policy (100% crash), yielding high aggregate crash rates of 60.7% and 80.0% respectively. This bimodal collapse—present in the majority of PPO seeds evaluated under the base config after aggressive training—is consistent with PPO's on-policy updates polarising toward local optima rather than learning a robust mixed strategy.

C. Stable-Baselines3 cross-validation

Because SB3 uses an `MlpPolicy` $[64, 64]$ trunk, its closest architectural counterpart is our custom **PPO+MLP** (223.0 ± 19.6), not the attention model. The SB3 run (seed 0, hyperparameters matched to our implementation) reached a final rolling-average reward of **260.5** and a peak of **266.7**. This single-seed value exceeds *every* one of our custom PPO seeds (best individual: PPO+Attention seed 1, 255.3; best PPO+MLP seed, 248.4) and sits ≈17% above our PPO+MLP 5-seed mean. We read this gap not as evidence that our PPO is broken—our implementation reaches the same peak (266.7) and the same qualitative learning dynamics (rapid rise before ≈10k steps, then a high plateau)—but as a reflection of SB3's mature engineering defaults (orthogonal initialisation, running observation normalisation, tuned optimiser ϵ) that we did not all replicate. Because SB3 is a single seed against our five, this is a directional sanity check on correctness, not a like-for-like performance claim; a fully fair comparison would require multi-seed SB3 runs with the identical feature extractor.

D. Reward-shaping study

Beyond the core ablation we trained PPO+Attention under the *conservative* and *base* reward styles (seeds 0, 1, and 2; $n = 3$) to illustrate how the **reward specification dominates qualitative behaviour**. Table V shows the final training reward per style.

The apparent ordering (conservative > base > aggressive in absolute reward) does **not** imply conservative is the "best" policy. The reward functions differ fundamentally: conservative targets low speed ($[10, 20]$ m/s), penalises lane changes (-0.5), and incurs a smaller collision penalty (-2.0), incentivising

TABLE VI

REWARD-SHAPING BEHAVIOURAL COMPARISON (PPO+ATTENTION, 30 EPISODES, BASE TEST CONFIG, SEEDS 0–2 AVERAGED FOR CONSERVATIVE AND BASE ($n = 3$); SEEDS 0–4 AVERAGED FOR AGGRESSIVE ($n = 5$)). HIGHER SURVIVAL / LOWER CRASH INDICATES SAFER DRIVING.

Training Style	Crash (%)	Survival	Spd (m/s)	Lane chg.
Conservative	$33.3^{\dagger}$	293.3	21.67	26.64
Base	**0.0**	400.0	20.03	0.0
Aggressive	$60.7^{\dagger}$	199.9	22.99	25.23

†Bimodal. Conservative: seeds 0,1 safe (0% crash, 400 steps), seed 2 catastrophic (100% crash, 80 steps). Aggressive: seeds 0,3 safe ($\leq$3.3%), seeds 1,2,4 catastrophic (100% crash).

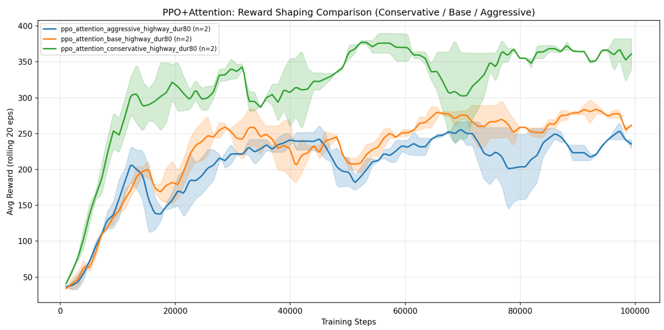

Fig. 2. PPO+Attention learning curves under three reward styles (conservative, base, aggressive). Returns are *not* comparable across styles due to differing reward coefficients (see Table I).

slow, stable driving that accumulates many small rewards over the full 400-step horizon. Aggressive targets high speed ($[30, 40]$ m/s), rewards lane changes ($+0.2$), and imposes a heavy collision penalty (-4.0); a single crash can wipe out many steps of reward, depressing absolute return. The comparison is *meaningless* across styles; the point of this study is the *behavioural* contrast revealed by Table VI.

Even under the same base test configuration, models trained under different reward styles drive markedly differently (Table VI): the base-trained policy is uniformly safe (0% crash), the aggressive-trained policy is fastest but crashes most, and the conservative-trained policy splits bimodally across seeds (the per-seed breakdown is examined in Sec. VII). These differences arise from the reward specification rather than the architecture, confirming reward design as the primary lever for qualitative driving style.

VII. DISCUSSION

Reward parity vs. behavioural divergence. The core finding of the ablation is that a parameter-matched attention architecture, a plain MLP, and a simpler DQN all converge to statistically comparable final episodic returns (223–235, a maximum gap of 5.7%). However, this near-parity in aggregate reward masks profound differences in the learned policies. As shown in Table IV, despite similar training returns, DQN achieves a robust lane-management strategy with an 11.3% crash rate, while PPO variants collapse into bimodal extremes yielding aggregate crash rates exceeding 60%. This serves as a critical lesson in applied reinforcement learning: **episodic return alone is an insufficient metric for safety-critical tasks**. High-variance policies can exploit the reward function to achieve comparable mean scores while exhibiting unacceptable physical behaviours—a failure mode that only multi-seed behavioural evaluation can expose.

The stabilising effect of off-policy replay. Contrary to expectations that a more complex feature extractor (Self-Attention) would inherently yield more stable control, Double DQN emerges as both the most stable learner ($\sigma = 7.7$) and the lowest-crash policy at deployment (subject to the swerving caveat of Sec. VI-B). We attribute the *training stability* to the fundamental difference in how experience is utilised. In the collision-heavy early learning phase, PPO's on-policy updates are highly susceptible to the non-stationarity of recent rollouts, leading some seeds to polarise toward catastrophic local optima (e.g. maintaining maximum speed regardless of obstacles). DQN's experience-replay buffer acts as a stabilising mechanism, maintaining a diverse distribution of transitions that prevents the Q-network from overfitting to short-term catastrophic sequences. The attention mechanism provides only a marginal variance reduction over a plain MLP for PPO (16.7 vs. 19.6), suggesting that the off-policy replay mechanism is a far more dominant factor in policy stability than the inductive bias of the neural architecture.

Sample efficiency: why DQN learns faster initially. Double DQN reaches rolling-avg reward $\geq$150 at a median of $\approx$7,721 environment steps, roughly **25%** earlier than PPO+Attention ($\approx$10,240) and **16%** earlier than PPO+MLP ($\approx$9,216). This aligns with a well-known property of off-policy learning: experience replay allows DQN to reuse early collision-heavy transitions many times, smoothing the otherwise high-variance gradient signal in the early training phase. PPO, which can only learn from its most-recent rollout, must accumulate sufficient on-policy experience before gradients stabilise. However, the early-learning advantage of DQN does not persist to convergence: all methods ultimately reach similar final rewards ($\sim$223–235).

A caveat on causality: train/test distribution shift. All models were trained under the *aggressive* style but evaluated under the *base* style—an intentional distribution shift that places every policy on a common yardstick. This design carries a confound we state plainly: DQN's lower crash rate may reflect that its policy *transfers* better across reward distributions, rather than that off-policy replay confers *intrinsic* safety. We therefore present the replay-buffer explanation as a plausible *association*, not a proven cause. Cleanly separating the two would require evaluating each model under its own (aggressive) training distribution as well; this is the most immediate follow-up. What the present data *do* support unambiguously is the headline point: near-tied episodic returns (223–235) can hide an order-of-magnitude difference in deployed crash rate (11.3% vs. 60–80%).

Reward shaping dominates—but variance qualifies the claim. Table VI reveals a nuanced picture. The base-trained policy (tested under its own training distribution) achieves a crash rate of only 0.0% and survives 400.0 steps on average with zero lane changes, the most consistent and safest profile of all three styles. The conservative-trained policy, which *intended* to be even safer through a stricter lane-change penalty (-0.5) and lower speed target ($[10, 20]$ m/s), shows a bimodal outcome: seeds 0 and 1 achieve 0% crash (400 steps, no lane changes, as intended), while seed 2 crashes on every single episode (100% crash rate, 80 lane changes per episode). The aggregate mean (33.3% crash) is therefore misleading—it reflects a high variance in policy convergence rather than an inherently mediocre strategy.

This finding *qualifies* but does not overthrow the reward-shaping dominance conclusion: reward engineering does constrain the *direction* of policy learning (conservative configurations tend toward lower-speed, fewer lane-change optima), but cannot guarantee that all random seeds converge to the intended behavioural basin. In combination with the high variance already documented in Table III ($\sigma = 16.7$ for PPO+Attention), this suggests that PPO on this benchmark has a multi-modal loss landscape, and some seeds get trapped in local optima that are high-reward under the training distribution but unsafe under slightly different test conditions. Reward engineering remains the *primary* design lever, but practitioners should validate with multi-seed behavioural evaluation—not just aggregate reward.

Implementation correctness. Our custom PPO reaches the same reward ceiling (266.7 peak) and the same qualitative learning dynamics as the SB3 reference, supporting the correctness of our GAE, clipped surrogate, advantage normalisation, and linear LR decay. We are deliberately cautious here: SB3's single-seed 260.5 exceeds all of our PPO seeds, so we claim only that our implementation is *in the same regime*, not that it matches SB3's tuned defaults. A definitive correctness check would require multi-seed SB3 runs with our exact architecture.

VIII. Limitations

- **Moderate seed count ($n = 5$).** Five seeds provide reasonable variance estimates but are insufficient for rigorous statistical significance claims; a formal test (e.g. Mann-Whitney U) requires more runs. We therefore phrase comparative results as "observed under these conditions."
- **Single primary environment.** The core ablation is on `highway-v0`. Multi-environment *support* is provided in the code, but generalisation of the ablation result across `merge`, `roundabout`, and `intersection` is not claimed.
- **Compute-bounded budget.** 100k steps is modest; some configurations may not be fully converged. Hyperparameters were matched, not individually tuned.
- **DQN is intentionally a simplified baseline** (smaller net, no attention variant, aggressive ϵ-decay); it bounds, rather than maximises, off-policy performance. It is also *not* capacity-matched to the PPO networks, so PPO-vs-DQN is an algorithm-family comparison, not a controlled-capacity one.
- **Train/test distribution shift.** Models are trained under the aggressive style and evaluated under the base style. This gives a common yardstick but confounds *intrinsic* safety with *transfer*: DQN's lower crash rate may partly reflect better cross-distribution transfer. The behavioural causal claims are therefore associational.

IX. Conclusion

We presented a from-scratch reproduction and parameter-matched architectural ablation of PPO and Double DQN on the discrete-action `highway-env` driving simulator ($n = 5$ seeds per configuration). Three key findings emerge:

(1) Return parity across architecture and algorithm. Under the aggressive reward style on `highway-v0`, all three configurations—PPO with Self-Attention (235.7 ± 16.7), PPO with MLP (223.0 ± 19.6), and Double DQN (234.1 ± 7.7)—approach the same theoretical reward ceiling (≈266.7), with final means differing by at most 5.7%. Architecture and algorithm family barely affect *how high* the agent scores—which is exactly what makes episodic return an unreliable basis for choosing between them.

(2) Return parity hides an order-of-magnitude behavioural gap. Despite near-tied returns, behavioural evaluation under a common base test configuration reveals crash rates of 11.3% (DQN) versus 60.7% (PPO+Attention) and 80.0% (PPO+MLP). DQN is also the most stable learner ($\sigma = 7.7$) and the fastest to reach avg≥150 (median ≈7,721 steps). We attribute DQN's *training* stability to experience replay smoothing gradient variance; its lower crash rate, however, we report cautiously—it is partly confounded by train/test distribution shift and by an unpenalised swerving strategy (Sec. VI-B), and is best read as an association, not proof of intrinsic safety.

(3) Reward shaping as primary lever—with an important variance caveat. A conservative-vs.-base-vs.-aggressive reward-shaping study confirms that reward specification is the dominant determinant of driving style. However, behavioural evaluation reveals a nuanced finding: even with the same reward shaping, individual seeds can converge to qualitatively opposite policies (e.g. conservative seeds 0,1: 0% crash; seed 2: 100% crash). Reward engineering constrains the *direction* of learning but does not guarantee all seeds converge to the intended behavioural basin. Multi-seed behavioural evaluation is therefore a necessary complement to reward design for safety-critical deployment.

Future work should evaluate each policy under its *own* (aggressive) training distribution to disentangle transfer from intrinsic safety, extend the ablation to `merge-v0` and `roundabout-v0` for environment generalisation, and test a prioritised-replay DQN variant to probe whether the stability advantage strengthens.

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